RESEARCH PLAN

Title: WaQuaCast: Development of an IoT-Based Machine Learning-Integrated Multi-Parameter Water Quality Monitoring and Forecasting System for the Balutakay River, Davao del Sur

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A. Rationale

Water quality is essential for maintaining public health, agricultural activities, and aquatic ecosystems. However, rivers such as the Balutakay River in Davao del Sur may experience changes in water quality due to agricultural runoff, weather conditions, and human activities. Conventional water-quality monitoring commonly relies on periodic water sampling and laboratory analysis, which can be time-consuming and may not provide immediate information about changes in water conditions. To address these challenges, this study aims to develop an IoT-based machine learning-integrated multi-parameter water-quality monitoring and forecasting system for the Balutakay River. The system will automatically collect measurements of pH, temperature, turbidity, and Total Dissolved Solids (TDS) and use the Weighted Arithmetic Water Quality Index (WAWQI) to assess and classify overall water quality. It will also use Random Forest Regression to analyze historical measurements and forecast future values of the selected parameters. Through the integration of IoT and machine learning, the study aims to provide a practical and data-driven approach that can support timely monitoring and assessment of changes in the water quality of the Balutakay River.

B. Research Questions

1. How accurately can the developed system measure the selected water-quality parameters compared with reference measurements?
    1.1. pH;
    1.2. Temperature;
    1.3. Turbidity; and
    1.4. Total Dissolved Solids (TDS)?
2. What is the water-quality condition of the selected monitoring sites based on the Weighted Arithmetic Water Quality Index (WAWQI)?
    2.1. Upstream site;
    2.2. Midstream site; and
    2.3. Downstream site?
3. How accurately can the Random Forest Regression model forecast the future values of the selected water-quality parameters using historical sensor data?
    3.1. pH;
    3.2. Temperature;
    3.3. Turbidity; and
    3.4. Total Dissolved Solids (TDS)?
4. Is there a significant difference between the actual and forecasted values of the selected water-quality parameters?
5. Which among the selected water-quality parameters can be forecasted most accurately by the Random Forest Regression model?

C. Goals and Expected Outcomes

1. To develop an IoT-based machine-learning water-quality monitoring system capable of measuring and forecasting pH, temperature, turbidity, and total dissolved solids (TDS) in selected sites along the Balutakay River, Davao del Sur.

2. To determine the accuracy of the developed system in measuring the selected water-quality parameters, namely pH, temperature, turbidity, and TDS.

3. To assess and classify the overall water quality of the selected monitoring sites using the Weighted Arithmetic Water Quality Index (WAWQI) based on the measured pH, temperature, turbidity, and TDS values and applicable DENR water-quality standards.

4. To evaluate the accuracy of the machine-learning model in forecasting the future values of pH, temperature, turbidity, and TDS using historical water-quality data. 5. To determine which of the four water-quality parameters can be forecasted most accurately by the developed machine-learning model.

D. Hypotheses

1.1 The machine-learning model cannot accurately forecast the future values of pH, temperature, turbidity, and total dissolved solids (TDS) based on historical water-quality data collected

1.2 The developed IoT-based monitoring system cannot accurately output a scale-based water-quality classification using the Weighted Arithmetic Water Quality Index (WAWQI) and DENR DAO 2026-08 standards compared with established reference classifications.

2.1 The machine-learning model can accurately forecast the future values of pH, temperature, turbidity, and total dissolved solids (TDS) based on historical water-quality data collected

2.2 The developed IoT-based monitoring system can accurately output a scale-based water-quality classification using the Weighted Arithmetic Water Quality Index (WAWQI) and DENR DAO 2026-08 standards compared with established reference classifications.

E. Description in Details and Methods or Procedure

Phase I

The researchers will design and develop an IoT-based machine learning-integrated multi-parameter water-quality monitoring and forecasting system for selected sites along the Balutakay River, Davao del Sur. The system will consist of pH, temperature, turbidity, and TDS sensors connected to an ESP32, Raspberry Pi, display, power supply, and other necessary electronic components. The ESP32 will collect and transmit the sensor measurements, while the Raspberry Pi will receive, store, and process the collected data. The system will be programmed to automatically record the measurements at predetermined time intervals for the development of a historical water-quality dataset. The use of IoT-based systems for continuous and automated water-quality monitoring has been demonstrated in previous studies (Abrajano et al., 2024; Krklješ et al., 2024).

The sensors will be calibrated using appropriate standard solutions or reference instruments before deployment to improve the reliability and accuracy of the measurements. Calibration results will be recorded and used to identify possible measurement errors. The developed system will then undergo preliminary testing to ensure that the sensors, data-processing components, storage system, and communication between the ESP32 and Raspberry Pi are functioning properly (U.S. Environmental Protection Agency [EPA], 2009).

Phase II

Water-quality measurements will be collected from three selected monitoring sites along the Balutakay River, Davao del Sur, representing the upstream, midstream, and

downstream sections of the river. The upstream site will be selected in a relatively less disturbed portion of the river and will serve as a reference point for the observed water-quality conditions. The midstream site will be located in an area influenced by nearby communities, agricultural activities, or other human activities. The downstream site will be located farther along the river, after areas with potential human and environmental influences, to assess the accumulated changes in water quality. The exact locations will be determined through preliminary site reconnaissance based on accessibility, safety, surrounding land use, potential sources of runoff or discharge, and suitability for sensor deployment.

At each monitoring site, the developed system will be positioned appropriately to allow the pH, temperature, turbidity, and TDS sensors to measure the water conditions. The sensors will be immersed in the water according to their required operating conditions, and measurements will be allowed to stabilize before being recorded. The system will automatically collect and store the measurements together with the corresponding date, time, and monitoring site. Multiple measurements will be obtained during the designated monitoring period to establish a sufficient historical dataset for water-quality assessment and machine-learning forecasting.

For sensor accuracy validation, measurements obtained from the developed system will be compared with appropriate calibrated reference instruments or established measurement procedures. Where applicable, water samples may also be collected from the three monitoring sites for comparison with laboratory or reference measurements. The differences between the sensor readings and reference measurements will be determined to evaluate the accuracy and reliability of each sensor (U.S. EPA, 2009).

Phase III

The collected measurements of pH, temperature, turbidity, and total dissolved solids (TDS) will be processed and used to determine the overall water quality through the

Weighted Arithmetic Water Quality Index (WAWQI). The selection and interpretation of these physicochemical parameters will be referenced to the applicable water-quality standards established by the Department of Environment and Natural Resources (DENR), specifically DAO 2016-08 . The WAWQI will then be calculated by assigning appropriate quality ratings and weights to the selected parameters based on the established methodology (Brown et al., 1970).

The calculated WAWQI value will then be used to determine the corresponding water-quality classification of the monitored river. The classification results will be recorded for each monitoring period and selected site to identify changes in the overall water-quality condition.

Phase IV

The historical measurements collected by the monitoring system will be organized into a dataset for machine-learning development. The dataset will contain the recorded values of pH, temperature, turbidity, and TDS, together with their corresponding timestamps. The data will be cleaned and checked for missing, inconsistent, or abnormal values before being used for model development.

The study will use Random Forest Regression as the machine-learning algorithm. The model will be trained using historical water-quality measurements to identify patterns and relationships that can be used to forecast future values of pH, temperature, turbidity, and TDS. Random Forest is an ensemble learning method that combines multiple decision trees to improve predictive performance (Breiman, 2001). The dataset will be divided into training and testing sets while maintaining the chronological order of the observations to avoid using future information when training the model.

After training, the Random Forest Regression model will generate forecasts for the selected water-quality parameters. The predicted values will then be compared with the actual

recorded values from the testing dataset to determine the model's forecasting performance (Hyndman & Athanasopoulos, 2021).

Phase V

The performance of the Random Forest Regression model will be evaluated separately for pH, temperature, turbidity, and TDS using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and coefficient of determination (R^2). These measures are commonly used to evaluate the performance of predictive and forecasting models (Hyndman & Athanasopoulos, 2021; James et al., 2021).

MAE will be used to determine the average magnitude of the forecasting errors, while RMSE will be used to determine the magnitude of prediction errors while giving greater influence to larger errors. R^2 will be used to determine how well the model explains the variation in the actual water-quality measurements. The parameter with the lowest forecasting errors and highest R^2 will be identified as the parameter that the model forecasts most accurately.

The final results will be used to evaluate whether the developed system can provide reliable water-quality measurements, DENR-referenced WAWQI-based classification, and future-value forecasts for the selected sites along the Balutakay River, Davao del Sur.

F. Risk and Safety

The study is expected to pose minimal risk to the researchers because it will primarily involve the operation of electronic components, water-quality sensors, and the collection of water samples from selected sites along the Balutakay River. Throughout the study, appropriate safety procedures will be followed to protect the researchers and prevent damage to the equipment. Personal protective equipment (PPE), including gloves, laboratory coats, and appropriate footwear, will be used during water sampling and sensor handling. Researchers will exercise caution when working near the river, particularly on wet, slippery,

or uneven surfaces, and will avoid entering unsafe areas or conducting sampling during hazardous weather conditions. Electrical components, batteries, and power supplies will be kept properly enclosed and protected from direct contact with water to prevent electrical hazards and equipment damage.

Before each field deployment, the monitoring system and its components will be inspected to ensure that there are no exposed wires, damaged cables, loose connections, or other conditions that may cause electrical hazards or equipment failure. The power supply and other electrical components will be kept properly enclosed and protected from water exposure. Water samples will also be collected using clean containers and handled carefully to prevent contamination that could affect the accuracy of the reference measurements. Any damaged electrical components, cables, or connectors will be removed from use and replaced or repaired before further operation. These measures will further reduce potential risks and help maintain the reliability of the data collected throughout the study.

During field activities, researchers will conduct sampling in groups and, when possible, inform a responsible person of the sampling schedule and location. The researchers will also monitor weather and river conditions before and during sampling and will immediately suspend field activities if conditions become unsafe. Care will be taken to minimize disturbance to the river and its surrounding environment, and no unnecessary materials or waste will be left at the sampling sites. These precautions will help ensure the safety of the researchers while maintaining responsible environmental practices throughout the study.

After data collection, water samples and other materials used during the study will be properly handled and disposed of in accordance with applicable environmental and laboratory safety guidelines. Electronic components and reusable equipment will be cleaned and maintained properly to prevent contamination and ensure safe operation. By following these

safety procedures, the study will minimize potential risks to the researchers, protect the monitoring equipment, and promote responsible environmental practices.

G. Data Analysis

The data obtained from the water-quality monitoring system will be analyzed using descriptive and predictive statistical methods to evaluate the accuracy of the sensors and the performance of the machine-learning model. The mean and standard deviation will be calculated for the measured pH, temperature, turbidity, and TDS values to describe the central tendency and variation of the collected data. These descriptive statistics are commonly used to summarize and describe the characteristics and variability of a dataset (Montgomery & Runger, 2018). The sensor measurements will also be compared with appropriate reference measurements to determine the accuracy and consistency of the developed monitoring system.

In addition, the Weighted Arithmetic Water Quality Index (WAWQI) will be used to integrate the measured water-quality parameters into a single numerical value representing the overall water quality. Based on the weighted arithmetic index approach developed by Brown et al., the method assigns weights and quality ratings to the selected water-quality parameters and combines them to obtain an overall water-quality index (Brown et al., 1970). Recent studies have continued to apply this approach to river and stream water-quality assessment (Fashina et al., 2023; Velasco et al., 2025). The resulting WAWQI values will be used to classify the water quality at the selected sampling sites and provide a reference classification for evaluating the machine-learning model.

For the machine-learning component, the collected historical data will be divided into training and testing datasets. The training dataset will be used to develop the machine-learning model, while the testing dataset will be used to evaluate its performance on

previously unseen data. Separating data for training and testing helps assess whether a predictive model can generalize beyond the data used to develop it (James et al., 2021).

The forecasting performance of the machine-learning model will be evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). MAE measures the average magnitude of prediction errors, while RMSE gives greater weight to larger errors. R^2 indicates the proportion of variation in the observed values that is explained by the model (Hyndman & Athanasopoulos, 2021). Lower MAE and RMSE values will indicate smaller forecasting errors, while a higher R^2 value will indicate better agreement between the predicted and actual measurements.

The forecasting results for pH, temperature, turbidity, and TDS will be analyzed separately to determine which parameter can be forecasted most accurately. The results will then be used to evaluate whether the developed machine-learning-integrated monitoring system can provide reliable forecasts of changes in the selected water-quality parameters.

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